

RoboCupJunior Soccer Entry Rules 2025

1:1 Standard Kit League & 1:1 Lightweight League

Soccer League Committee 2025:

Hikaru Sugiura	USA
Jakub Gál	Slovakia
Mohammad Hadi Shirani	Iran
David Schwarz	Germany
William Plummer	Australia (CHAIR)
Isa El Doori	Netherlands

Soccer League Committee 2024:

Michael Ambrose	USA
Ryely Burtenshaw-Day	New Zealand
Ivan Kolarić	Croatia
David Schwarz	Germany
William Plummer	Australia (CHAIR)
Adrián Matejov	Slovakia



Regional and super-regional tournament organizers may use these rules, make changes specific to their regions or not run Entry leagues at all. Contact your regions or super-regions tournament organizers to find out which rules are going to be played at your tournament.



This ruleset is not intended to be used at international level, except for super-regional tournaments.

These are the RoboCupJunior Soccer Entry rules for the **1:1 Lightweight League** and **1:1 Standard Kit League** proposed for suggested use by regional and super-regional tournaments in the 2025 season. They are released by the RoboCupJunior League Committee. The English version of these rules has priority over any translations.

The aim of this document is to provide entry-level rulesets for RoboCupJunior Soccer that are harmonized across regions and that may be used as-is or adapted to specific needs at regional and super-regional tournaments. Some regions already have their own, separate version of Soccer Entry rules. Teams are advised to check with the local tournament organizers and Regional Representative regarding updates and changes to this ruleset specific to their location. Each team is responsible for verifying the correct and latest version of the rules prior to competition.

Figure 1 Two teams with one LWL robot each will compete using an IR ball on RCJ Soccer fields without the out-area. There is no need for using camera vision or line detection. Photo: Andreas Lander



Preface

In the RoboCupJunior Soccer Entry challenge, teams of young engineers design, build, and program one fully autonomous mobile robot to compete against another team in matches. The robots must detect a ball and score into a color-coded goal on a special field that resembles a human soccer field.

To be successful, participants must demonstrate skill in programming, robotics, electronics and mechanics. Teams are also expected to contribute to the advancement of the community as a whole by sharing their discoveries with other participants and by engaging in good sportsmanship, regardless of culture, age or result in the competition. **All are expected to compete, learn, have fun, and grow.**

RoboCupJunior Soccer Entry rules are supplied for two entry-level. **1:1 Lightweight League** leagues modelled after LWL but reduced to one robot and restricted to at most three driving motors and **1:1 Standard Kit League** further restricted to two driving motors and restricted to a list of similarly powered motors (see Appendix C, [Motor Whitelist](#)).

1 Differences to 2:2 League rulesets

1.0.1

These rules are based on the 2:2 rulesets. Their main differences compared to those are:

1.1 Both 1:1 Leagues

- **The Field** has no out of bounds areas and gameplay may touch the walls. This reduces the complexity of robot construction and gameplay. Use of many existing fields is possible with no or minor modifications due to flexible dimensions.
- Each team starts **only one robot**
- **The Ball** the league uses the same special ball that emits an IR signal that Lightweight League uses. Please see Rule 6, [BALL](#) for balls specifications.
- Restrictions on the number of drive motors apply (max. 2 drive motors for **1:1 Standard Kit League**, max. 3 drive motors for **1:1 Lightweight League**)

- In most regions participation in each 1:1 League is limited to two years (see Rule 1.4, Competition Cap)
- **Furthermore, robots that have been constructed for any of the 2:2 RobocupJunior Soccer leagues at any level (local, regional, super-regional, international) in past, future, or present may not be used at any 1:1 League match.**

1.2 1:1 Standard Kit League

- Number of drive motors is restricted to two
- Due to availability concerns with sensors and future products suitable for RoboCupJunior Soccer use the restrictions to only Lego and Fischertechnik parts were lifted. A number of changes were made to preserve the character of the league and keep existing Lego and Fischertechnik robots competitive:
- Use of parts that are neither Lego nor Fischertechnik is permitted
- **A weight limit of 1400g is introduced**
- Only a list of allowable motors (see Appendix C, Motor Whitelist) with comparable power to Lego and Fischertechnik motors is permitted

1.3 1:1 Lightweight League

- Number of drive motors is restricted to three
- The voltage limit remains at 12V (unlike 2:2 LWL where it is increased)
- **The weight limit for 1:1 Lightweight League has been increased to match the 2:2 LWL level of 1400g**

Changes from the 2024 RoboCupJunior Soccer Entry Rules

- Added "Furthermore, robots that have been constructed for any of the 2:2 RobocupJunior Soccer leagues at any level (local, regional, super-regional, international) in past, future, or present may not be used at any 1:1 League match."
- Added "A weight limit of 1400g is introduced"
- Added "The weight limit for 1:1 Lightweight League has been increased to match the 2:2 LWL level of 1400g"
- Added "QITA"
- Added "Vex Motor"
- Added "TT Motor"

1.4 Competition Cap

1.4.1

Team members can participate only twice in each league played according to one of these RoboCupJunior Soccer Entry rulesets. After their second participation, they need to move on from **1:1 Standard Kit League** to **1:1 Lightweight League** and from **1:1 Lightweight League** to one of the 2:2 Leagues respectively. ¹

¹ Some regions will have other entry leagues that may be exempt from this rule. Please refer to your local tournament organizers in order to find out in which leagues you may compete.

- 1.4.2 Team members that participated in the 2:2 RoboCupJunior Soccer leagues at any level (local, regional, super-regional, international) before may not participate in the Entry league(s) again.

Construction and Programming have to be performed exclusively by the students

- 1.4.3 Robots must be constructed and programmed exclusively by student members of the team. Mentors, teachers, parents or companies should not be involved in the design, construction, assembly, programming or debugging of robots. To avoid possible disqualification, it is extremely important that teams abide by these leagues' regulations, especially Rule 8.5.F, **Construction** and Rule 8.5.G, **Programming**, and all other competitor's rules.
- 1.4.4 If in doubt, please consult with your Regional Representative before registering your team.

2 GAMEPLAY

2.1 Game procedure and length of a game

- 2.1.1 RCJ Soccer games consist of two teams of one robot each, playing soccer against each other. Each team has one autonomous robot. The game will consist of two halves. The duration of each half is 10-minutes. There will be a 5-minute break in between the halves.
- 2.1.2 The game clock will run for the duration of the halves without stopping (except when a referee wants to consult another official). The game clock will be run by a referee or a referee assistant (see Rule 8.1, **Referee and referee assistant** for more information on their roles).
- 2.1.3 Teams are expected to be at the field 5 minutes before their game starts. Being at the inspection table does not count in favor of this time limit. Teams that are late for the start of the game may be penalized one goal **per 30 seconds** at the referee's discretion.
- 2.1.4 The final game score will be trimmed so that there is at most 10-goal difference between the losing and the winning team.

2.2 Pre-match meeting

- 2.2.1 At the start of the first half of the game, a referee will toss a coin. The team mentioned first in the draw shall call the coin. The winner of the toss can choose either which end to kick towards, or to kick off first. The loser of the toss chooses the other option. After the first half, teams switch sides. The team not kicking off in the first half of the game will kick off to begin the second half of the game.
- 2.2.2 During the pre-match meeting the referee or their assistant may check whether the robots are capable of playing (i.e., whether they are at least able to follow and react to the ball). If none of the robots is capable of playing, the game will not be played and zero goals will be awarded to both teams.

2.3 Kick-off

- 2.3.1 Each half of the game begins with a kick-off. All robots must be located on their own side of the field. All robots must be halted. The ball is positioned by a referee in the center of the field.
- 2.3.2 The team kicking off places their robot on the field first.
- 2.3.3 The team not kicking off will now place their robot on the defensive end of the field. The robot on the team not kicking off must be at least 30 cm away from the ball (outside of the center circle).

- 2.3.4 Robots cannot be placed inside the goal. Robots cannot be repositioned once they have been placed, except if the referee requests to adjust their placement to make sure that the robots are placed properly within the field positions.
- 2.3.5 On the referee's command (usually by whistle), all robots will be started immediately by each captain. Any robots that are started early will be removed by the referee from the field and deemed damaged.
- 2.3.6 Before a kick-off, **all damaged robots** are allowed to return to the playing field immediately if they are *ready and fully functional*.
- 2.3.7 If no robots are present at a kick-off (because they are damaged Rule 2.8, **Damaged robots**), the penalties are discarded and the match resumes with a Rule 2.3.A, **Neutral kick-off**.

2.3.A Neutral kick-off

- 2.3.A.1 A neutral kick-off is the same as the one described in Rule 2.3, **Kick-off** with a small change: all robots must be at least 30 cm away from the ball (outside of the center circle).

2.4 Human interference

- 2.4.1 Except for the kick-off, human interference from the teams (e.g. touching the robots) during the game is not allowed unless explicitly permitted by a referee. Violating team(s)/team member(s) may be disqualified from the game.
- 2.4.2 The referee or a referee assistant can help robots get unstuck if the ball is not being disputed near them and if the situation was created from normal interaction between robots (i.e. it was not a design or programming flaw of the robot alone). The referee or a referee assistant will pull back the robots just enough for them to be able to move freely again.

2.5 Ball movement

- 2.5.1 A robot cannot hold a ball. Holding a ball is defined as taking full control of the ball by removing all of degrees of freedom. Examples for ball holding include fixing a ball to the robot's body, surrounding a ball using the robot's body to prevent access by others, encircling the ball or somehow trapping the ball with any part of the robot's body. If a ball does not roll while a robot is moving, it is a good indication that the ball is trapped.
- 2.5.2 The only exception to holding is the use of a rotating drum (a "dribbler") that imparts dynamic back spin on the ball to keep the ball on its surface.
- 2.5.3 Other players must be able to access the ball.
- 2.5.4 The ball needs to stay within the bounds of the field, as defined by the walls. If a robot moves the ball outside of the field (that is, beyond the walls or above their height), it is deemed damaged. (Rule 2.8, **Damaged robots**)

2.6 Scoring

- 2.6.1 A goal is scored when the ball strikes or touches the back wall of the goal. Goals scored by any robot have the same end result: they give one goal to the team on the opposite side. After a goal, the game will be restarted with a kick-off from the team who was scored against.

2.7 Lack of progress

- 2.7.1 Lack of progress occurs if there is no progress in the gameplay for a reasonable period of time and the situation is not likely to change. Typical lack of progress situations are when the ball is stuck between robots, when there is no change in ball and robot's positions, or when the ball is beyond detection or reach capability of all robots on the field.
- 2.7.2 After a visible and loud count ², a referee will call **lack of progress** and will move the ball to the nearest unoccupied neutral spot. If this does not solve the lack of progress, the referee can move the ball to a different neutral spot.

2.8 Damaged robots

- 2.8.1 If a robot is damaged, it has to be taken off the field and must be fixed before it can play again. Even if repaired, the robot must remain off the field for at least one minute or until the next kick-off is due.
- 2.8.2 Some examples of a damaged robot include:
- it does not respond to the ball, or is unable to move (it lost pieces, power, etc.).
 - it turns over on its own accord.
- 2.8.3 Computers and repair equipment are not permitted in the playing area during gameplay. Usually, a team member will need to take the damaged robot to an "approved repair table" near the playing area. A referee may permit robot sensor calibration, computers and other tools in the playing area, only for the 5 minutes before the start of each half.
- 2.8.4 After a robot has been fixed, it will be placed on the unoccupied neutral spot furthest from the ball, facing its own goal. A robot can only be returned to the field if the damage has been repaired. If the referee notices that the robot was returned to the field with the same original problem, they may ask the robot to be removed and proceed with the game as if the robot had not been returned.
- 2.8.5 **Only the referee decides whether a robot is damaged.** A robot can only be taken off or returned with the referee's permission.
- 2.8.6 Whenever a robot is removed from play, its motors must be turned off.

2.9 Interruption of Game

- 2.9.1 In principle, a game will not be stopped.
- 2.9.2 A referee can stop the game if there is a situation on or around the field which the referee wants to discuss with an official of the tournament or if the ball malfunctions and a replacement is not readily available.
- 2.9.3 When the referee has stopped the game, all robots must be stopped and remain on the field untouched. The referee may decide whether the game will be continued/resumed from the situation in which the game was stopped or by a kick-off.

² usually a count of three

3 TEAM

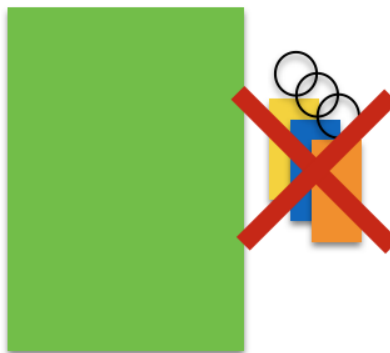
3.1 Regulations

- 3.1.1 A team must have more than one member to form a RoboCupJunior team to participate in the competition. Team member(s) and/or robot(s) cannot be shared between teams. The maximum number of team members is defined by each competition respectively, but is usually 4.
- 3.1.2 Each team member needs to carry a technical role.
- 3.1.3 Each team must have a **captain**. The captain is the person responsible for communication with referees. The team can replace its captain with another team member during the competition. Each team is allowed to have at most two members beside the field during gameplay: they will usually be the captain and an assistant team member.

3.2 Violations

- 3.2.1 Teams that do not abide by the rules are not allowed to participate.
- 3.2.2 Any person close to playing fields with walls lower than 22cm is not allowed to wear any yellow or blue clothes that can be seen by the robots (to avoid interference). A referee can require a team member to change clothes or to be replaced by another team member if interference is suspected.
- 3.2.3 The referee can interrupt a game in progress if any kind of interference from spectators is suspected (color clothing, IR emitters, camera flashes, mobile phones, radios, computers, etc.).
- 3.2.4 This needs to be confirmed by the tournament organizers if a claim is placed by the other team. A team claiming that their robot is affected by colors has to show the proof/evidence of the interference.

Figure 2 Anyone close to the playing field is not allowed to wear orange, yellow or blue clothes



4 ROBOTS

4.1 Number of robots / substitutions

- 4.1.1 Each team is allowed to have only one robot for the full tournament. The substitution of robots during the competition within the team or with other teams is forbidden.

4.2 Interference

- 4.2.1 Robots are not allowed to be colored yellow or blue in order to avoid interference. Yellow or blue colored parts used in the construction of the robot must either be occluded by other parts from the perception by another robot or be taped/painted with a neutral color.
- 4.2.2 Robots must not produce magnetic interference in another robot on the field.
- 4.2.3 Robots must not produce visible light that may prevent the opposing team from playing when placed on a flat surface. Any part of a robot that produces light that may interfere with the opposing robot's vision system must be covered.
- 4.2.4 A team claiming that their robot is affected by the other team's robot in any way must show the proof/evidence of the interference. Any interference needs to be confirmed by the tournament organizers if a claim is placed by the other team.

4.3 Control

- 4.3.1 The use of remote control of any kind is not allowed during the match. Robots must be started and stopped manually by humans and be controlled autonomously.

4.4 Agility

- 4.4.1 Robots must be constructed and programmed in a way that their movement is not limited to only one dimension (defined as a single axis, such as only moving in a straight line). They must move in all directions, for example by turning.
- 4.4.2 Robots must respond to the ball in a direct forward movement towards it.
- 4.4.3 A robot must touch the ball that is placed no further than 20 cm from it within 10 seconds. If a robot does not do so within the time limit, it is deemed to be damaged. (See [Damaged Robots](#).)
- 4.4.4 Robots may enter the goals.

4.5 Handle

- 4.5.1 All robots must have a stable and easily noticeable handle to hold and to lift them. The handle must be easily accessible and allow the robot to be picked up from at least 5 cm above the highest structure of the robot.
- 4.5.2 The dimensions of the handle may exceed the robot height limitation, but the part of the handle that exceeds this limit cannot be used to mount components of the robot.

4.6 Top Markers

- 4.6.1 Top markers (as required in the 2:2 leagues) are not required.

4.7 Violations

- 4.7.1 Robots that do not abide by these specifications/regulations are not allowed to play.
- 4.7.2 If violations are detected during a running game the team may be disqualified for that game.

- 4.7.3 If similar violations occur repeatedly, the team may be disqualified from the tournament.

5 FIELD

5.1 Dimensions of the field

- 5.1.1 The playing field is between 110 cm and 160 cm in width.
- 5.1.2 The playing field is between 180 cm and 225 cm in depth.
- 5.1.3 This allows for re-using existing equipment such as older RCJ soccer fields (122 cm by 183 cm, used to be called "Soccer A") or FLL (236 cm by 114 cm becoming 221 cm by 114 cm with temporary goals installed) or regular RCJ Soccer fields (219 cm x 158 cm when converted with temporary walls on field lines). Teams should contact the organizers of their tournament about the exact fields used for the competition.

5.2 Walls

- 5.2.1 Walls are placed all around the field. The height of the wall is between 10cm and 25cm. A height of at least 14 cm is recommended. The walls are painted matte black.
- 5.2.2 The four corners of the field are flattened in order to make it easier for robots to retrieve the ball from a corner. The flat area is approx. 14cm in width.
- 5.2.3 There is no outer area.

5.3 Goals

- 5.3.1 The field has two goals, centered on each of the shorter sides of the playing field. The goal inner space is 45 to 60 cm wide and 74 mm deep. It is outside the playing field (submerged into the walls). The height of the goal is equal to the height of the walls.
- 5.3.2 The goal **may or may not have** a cross-bar on top. The size of the cross-bar is 2 ± 1 cm in height.
- 5.3.3 The interior walls and the crossbar of each goal are colored matte, one goal yellow, the other goal blue.
- 5.3.4 It is recommended that the blue be of a brighter shade so that it is different enough from the black exterior.

5.4 Floor

- 5.4.1 The floor consists of green carpet ideally of darker shade on top of a hard level surface. Teams should be prepared to adjust to different levels of contrast between the green carpet and lines as some events may be restricted to using lighter shades of green. All lines on the field should be painted, marked with tape, or installed as white carpet and be somewhat resistant to tearing or ripping. Lines should have a width of 20mm ($\pm 10\%$).
- 5.4.2 It is impractical to set international constraints on carpet other than it being green. In the spirit of the competition, teams should design robots to be tolerant or adaptable to different fibers, textures, construction, density, shades, and designs of carpet especially when competing amongst different regions. Teams are encouraged to visit regional resources or reach out to Local Organization Committee for suggestions if desiring to build their own practice field(s).

5.5 Neutral spots

- 5.5.1 There are five neutral spots defined in the field. One is in the center of the field. The other four are adjacent to each corner, located 45 cm along the long edge of the field, aligned with each goal post towards the middle of the field (from the goal post). The neutral spots can be drawn with a thin black marker. The neutral spots ought to be of circular shape measuring 1 cm in diameter.

5.6 Center circle

- 5.6.1 A center circle will be drawn on the field. It is 60 cm in diameter. It is a thin black marker line. It is there for Referees and Captains as guidance during kick-off.

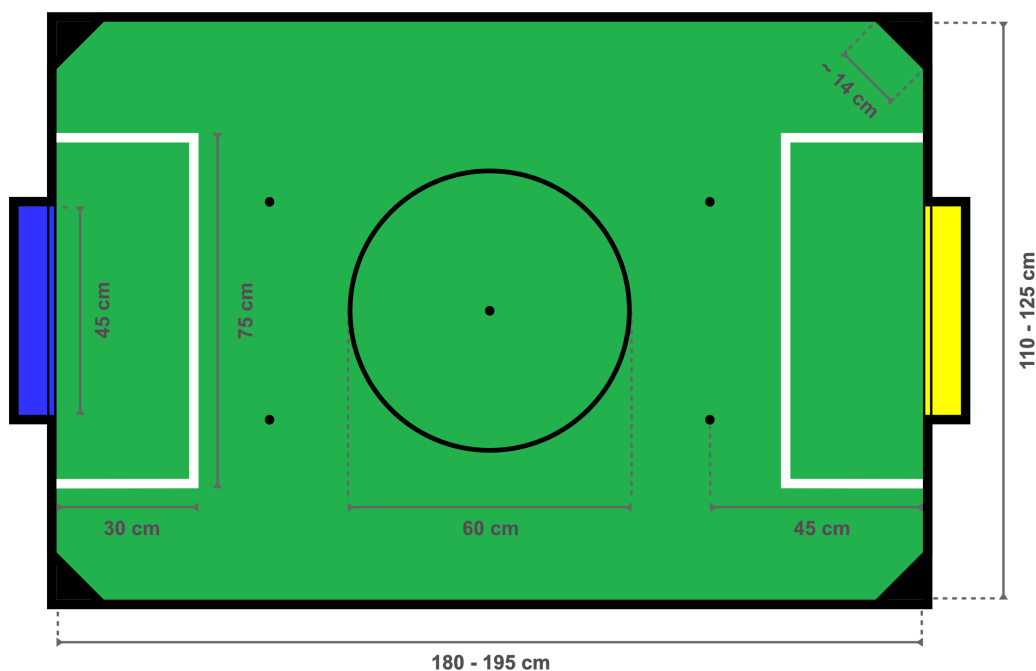
5.7 Penalty areas

- 5.7.1 In front of each goal there is a penalty area (optional). It is decorative and serves no purpose in gameplay.

5.8 Lighting and Magnetic Conditions

- 5.8.1 The tournament organizers will do their best to limit the amount of external lightning and magnetic interference. However, the robots need to be constructed in a way which allows them to work in conditions that are not perfect (i.e. by not relying on compass sensors or specific lightning conditions).

FIELD DIAGRAMS



- 5.8.2

6 BALL

6.1 Specification for Soccer Entry Ball

- 6.1.1 See Appendix A, [Technical Specification for pulsed Soccer Ball](#).

6.2 Tournament balls

- 6.2.1 Balls for the tournament must be made available by the tournament organizers. Tournament organizers are not responsible for providing balls for practice.

7 CODE OF CONDUCT

7.1 Fair Play

- 7.1.1 It is expected that the aim of all teams is to play a fair and clean game of robot soccer. It is expected that all robots will be built with consideration to other participants.
- 7.1.2 Robots are not allowed to cause deliberate interference with or damage to other robots during normal gameplay.
- 7.1.3 Robots are not allowed to cause damage to the field or to the ball during normal gameplay.
- 7.1.4 A robot that causes damage may be disqualified from a specific match at the tournament organizer's discretion.
- 7.1.5 Humans are not allowed to cause deliberate interference with robots or damage to the field or the ball.

7.2 Behavior

- 7.2.1 All participants are expected to behave themselves. All movement and behavior is to be of a subdued nature within the tournament venue.

7.3 Help

- 7.3.1 Mentors (teachers, parents, chaperones, and other adult team-members including translators) are not allowed in the student work area unless it is explicitly but temporarily permitted by tournament organizers. Only participating students are allowed to be inside the work area.
- 7.3.2 **Mentors must not touch, build, repair, or program any robots.**

7.4 Sharing

- 7.4.1 The understanding that any technological and curricular developments should be shared among the RoboCup and RoboCupJunior participants after the tournament has been a part of world RoboCup competitions.

7.5 Spirit

- 7.5.1 It is expected that all participants, students, mentors, and parents will respect the RoboCupJunior mission.
- 7.5.2 ***It is not whether you win or lose, but how much you learn that counts!***

7.6 Violations / Disqualification

- 7.6.1 Teams that violate the code of conduct may be disqualified from the tournament. It is also possible to disqualify only single person or single robot from further participation in the tournament.
- 7.6.2 In less severe cases of violations of the code of conduct, a team will be given a warning. In severe or repeated cases of violations of the code of conduct a team may be disqualified immediately without a warning.

8 CONFLICT RESOLUTION

8.1 Referee and referee assistant

- 8.1.1 The referee is a person in charge of making decisions with regards to the game, according to these rules, and may be assisted by a referee assistant.
- 8.1.2 **During gameplay, the decisions made by the referee and/or the referee assistant are final.**
- 8.1.3 Any argument with the referee or the referee assistant can result in a warning. If the argument continues or another argument occurs, this may result in immediate disqualification from the game.
- 8.1.4 Only the captain has a mandate to freely speak to the referee and/or their assistant. Shouting at a referee and/or their assistant, as well as demanding a change in ruling may be penalized by a warning at the referee's discretion.
- 8.1.5 At the conclusion of the game, the result recorded in the scoresheet is final. The referee will ask the captains to add written comments to the scoresheet if they consider them necessary. These comments will be reviewed by the tournament organizers.

8.2 Rule clarification

- 8.2.1 Rule clarification may be made by members of the tournament organizers and the Soccer League Committee, if necessary even during a tournament.

8.3 Rule modification

- 8.3.1 If special circumstances, such as unforeseen problems or capabilities of a robot occur, rules may be modified by the tournament organizers, if necessary even during a tournament.

8.4 Regulatory statutes

- 8.4.1 Each RoboCupJunior competition may have its own regulatory statutes to define the procedure of the tournament (for example the SuperTeam system, game modes, the inspection of robots, interviews, schedules, etc.). Regulatory statutes become a part of this rule.

8.5 Regulations

8.5.A Dimensions

8.5.A.1 Robots will be measured in an upright position with all parts extended. A robot's dimensions must not exceed the following limits:

sub-league	1:1 Standard Kit League	1:1 Lightweight League
size	22.4 cm (square) ^[0]	22.0 cm (circle) ^[1]
height	22.4 cm ^[2]	22.0 cm ^[2]
weight	1400 g	1400 g ^[3]
ball-capturing zone	3.0 cm	3.0 cm



[0] Robot must fit smoothly into a cube of this size.



[1] Robot must fit smoothly into a cylinder of this diameter



[2] The handle of a robot may exceed the height.



[3] The weight of the robot includes that of the handle.



[4] We **strongly** encourage teams to include protection circuits for Lithium-based batteries



[5] Voltage limits relate to the **nominal values**, slightly higher voltages at high states of charge are allowed.

- 8.5.A.2 Ball-capturing zone is defined as any internal space created when a straight edge is placed on the protruding points of a robot. This means the ball must not enter the convex hull of a robot by more than the specified depth. Furthermore, it must be possible for another robot to take possession of the ball.

8.5.B Infrared interference

- 8.5.B.1 Components designed to emit IR (e.g. ToF, LiDAR, IR distance sensors, IR LEDs/LASERs etc.) are not allowed and tournament organizers will require such devices to be removed or covered up.
- 8.5.B.2 Infrared light reflecting materials must not be visible. If robots are painted, they must be painted matte. Minor parts that reflect infrared light could be used as long as other robots are not affected.

8.5.C Limitations

- 8.5.C.1 A robot may use any number of cameras without restrictions on lenses, optical parts, optical systems, and total field of view. Components may be sourced in any way the team sees fit.
- 8.5.C.2 Pneumatic devices are allowed to use ambient air only.
- 8.5.C.3 Kicker strength is subject to compliance check at any time during the competition. During gameplay, a referee can ask to see a sample kick on the field before each half when a damaged robot is returned to the field or when the game is about to be restarted after a goal. If the referee strongly suspects that a kicker exceeds the power limit, they can require an official measurement. See Appendix B, [Kicker Power Measuring Procedures](#) for more details.

8.5.D 1:1 Standard Kit League

- 8.5.D.1 With exceptions mentioned below all parts may be used (the limitation to Lego and Fischertechnik parts only no longer applies).
- 8.5.D.2 The Soccer League Committee maintains a list of motors that are most likely allowed at all Entry tournaments. The tournament organizers will have the final say on this and may have a different list. Teams are expected to make sure their motors are allowed by checking the lists for their tournaments. This list is available Appendix C, [Motor Whitelist](#).
- 8.5.D.3 Voltage pump circuits are not permitted.
- 8.5.D.4 Maximum nominal battery voltage is 9V.
- 8.5.D.5 A maximum of two driving motors is allowed. Only motors involved in moving the robot across the field are considered driving motors. Any number of additional motors may be used for dribblers, kickers and other mechanisms.
- 8.5.D.6 A weight limit of 1400g is introduced to prevent heavy custom robots from overpowering Lego and Fischertechnik robots.

8.5.E 1:1 Lightweight League

- 8.5.E.1 Voltage pump circuits are permitted only for a kicker drive. No voltage may exceed 48V at any time and maximum boost voltage must be available for demonstration and measurement at inspections. When not in use measurement contacts must be protected from accidental touches or short circuits.
- 8.5.E.2 All other electrical circuits inside the robot cannot exceed 12.0 V. Each robot must be designed to allow verifying the voltage of power packs and its circuits, unless the nominal voltage is obvious by looking at

the robot, its power packs and connections.

- 8.5.E.3 The voltage limit remains 12V nominal and 48V maximum at the kicker despite changes in 2:2 Lightweight.
- 8.5.E.4 A maximum of three drive motors is allowed

8.5.F Construction



Robots must be constructed exclusively by the student members of a team. Mentors, teachers, parents or companies may not be involved in the design, construction, and assembly of robots.

- 8.5.F.1 For the construction of a robot, any robot kit or building block may be used as long as the design and construction are primarily and substantially the original work of a team. This means that commercial kits may be used but must be substantially modified by the team. It is neither allowed to mainly follow a construction manual, nor to just change unimportant parts.
- 8.5.F.2 Indications for violations are the use of commercial kits that can basically only be assembled in one way or the fact that robots from different team(s), build from the same commercial kit, all basically look or function the same.
- 8.5.F.3 Robots must be constructed in a way that they can be started by the captain without the help of another person.
- 8.5.F.4 Since a contact with an opponent robot and/or dribbler that might damage some parts of robots cannot be fully anticipated, **robots must have all its active elements properly protected with resistant materials**. For example, electrical circuits and pneumatic devices, such as pipelines and bottles, must be protected from all human contact and direct contact with other robots.



All driven dribbler gears must be covered with metal or hard plastic.

- 8.5.F.5 When batteries are transported or moved, it is **strongly** recommended that safety bags be used. Reasonable efforts should be made to make sure that in all circumstances robots avoid short-circuits and chemical or air leaks.



The use of swollen, tattered or otherwise dangerous battery is not allowed.

8.5.G Programming

- 8.5.G.1 Robots must be programmed exclusively by student members of the team. Mentors, teachers, parents or companies should not be involved in the programming and debugging of robots.
- 8.5.G.2 For the programming of the robots, any programming language, interface or integrated development environment (IDE) may be used. The use of programs that come together with a commercial kit (especially sample programs or presets) or substantial parts of such programs are not allowed. It is not allowed to use sample programs, not even if they are modified.

8.5.H Inspections

- 8.5.H.1 Robots must be inspected and certified every day before the first game is played. The tournament organizers may request other inspections if necessary, including random inspections which may happen at any time. The routine inspections include:
- Weight restrictions for the particular sub-league (see Rule 8.5.A, Dimensions).
 - Robot dimensions (see Rule 8.5.A, Dimensions).
 - Voltage restrictions (see Rule 8.5.A, Dimensions and Rule 8.5.C, Limitations).
 - Kicker strength limits, if the robot has a kicker (see Appendix B, Kicker Power Measuring Procedures).
- 8.5.H.2 Proof must be provided by each team that its robots comply with these regulations, for example, by a detailed documentation or logbook. Teams may be interviewed about their robots and the development process at any time during a tournament.

A Technical Specification for pulsed Soccer Ball

1.1 Preamble

- 1.1.1 Answering to the request for a soccer ball for RCJ tournaments that would be more robust to interfering lights, less energy consuming and mechanically more resistant, the Soccer League Committee defined the following technical specifications with the special collaboration from EK Japan and HiTechnic.
- 1.1.2 Producers of these balls must apply for a certification process upon which they can exhibit the RCJ-compliant label and their balls used in RCJ tournaments.
- 1.1.3 Balls with these specifications can be detected using specific sensors but also common IR remote control receivers (TSOP1140, TSOP31140, GP1UX511QS, etc. - on-off detection with a possible gross indication of distance).

1.2 Specifications

1.2.A IR light

- 1.2.A.1 The ball emits infra-red (IR) light of wavelengths in the range 920nm - 960nm, pulsed at a square-wave carrier frequency of 40 kHz. The ball should have enough ultra-bright, wide-angle LEDs to minimize unevenness of the IR output.

1.2.B Diameter

- 1.2.B.1 The diameter of the ball is required to be 74mm. A well-balanced ball shall be used.

1.2.C Drop Test

- 1.2.C.1 The ball must be able to resist normal gameplay. As an indication of its durability, it should be able to survive, undamaged, a free-fall from 1.5 meters onto a hardwood table or floor.

1.2.D Modulation

- 1.2.D.1 The 40 kHz carrier output of the ball shall be modulated with a trapezoidal (stepped) waveform of frequency 1.2 kHz. Each 833-microsecond cycle of the modulation waveform shall comprise 8 carrier pulses at full intensity, followed (in turn) by 4 carrier pulses at 1/4 of full intensity, four pulses at 1/16 of full intensity and four pulses at 1/64 of full intensity, followed by a space (i.e. zero intensity) of about 346 microseconds. The peak current level in the LEDs shall be within the range 45-55mA. The radiant intensity shall be more than 20mW/sr per LED.

1.2.E Battery Life

- 1.2.E.1 If the ball has an embedded rechargeable battery, when new and fully charged it should last for more than 3 hours of continuous use before the brightness of the LEDs drops to 90% of the initial value. If the ball uses replaceable batteries, a set of new high-quality alkaline batteries should last for more than 8 hours of continuous use before the brightness of the LEDs drops to 90% of the initial value.

1.2.F Coloration

- 1.2.F.1 The ball must not have any marks or discoloration that can be confused with goals, or the field itself.

1.3 Official suppliers for pulsed balls

- 1.3.1 Currently, there is one ball that has been approved by the Soccer League Committee:
- RoboCupJunior Soccer ball operating in MODE A (pulsed) made by EK Japan/Elekit (<https://elekit.co.jp/en/product/RCJ-05R>)
- 1.3.2 Note that this ball was previously called RCJ-05. While you may not be able to find a ball with this name anymore, any IR ball produced by EK Japan/Elekit is considered to be approved by the Soccer League Committee.

B Kicker Power Measuring Procedures

All robot kickers will be tested with the tournament ball. Kicker Power will be measured by means of an on-field test.

The test is performed as follows:

1. Place robot inside the left corner of a goal.
2. Perform a kick into the opposing goal
3. The kicker power test is passed if the ball does not cross into the opposite half of the field.

C Motor Whitelist

The below list of motors most likely be allowed in all 1:1 Standard Kit League tournaments. Teams are advised to check this document for updates (see date at the bottom of the page) **and** to check their local, regional and super-regional tournament regulations for the official list for the respective tournaments.

Manufacturer	Model Number	Notes
Lego	All models	Original and replica parts allowed
Fischertechnik	All models	Original and replica parts allowed
Pololu	Metal Gearmotor 25Dx48L mm MP 12V	with or without any of the gearbox ratios
Pololu	Metal Gearmotor 25Dx48L mm LP 12V	with or without any of the gearbox ratios
Robotis	Dynamixel XL-320, XL-330	
XYT	JGA25-370	
TT Motor	GMP16-050SH	
QITA	XYT 12V Mini DC Motor (JGA25-370)	
Vex Motor	IQ Smart Motor (228-2560)	
TT Motor	GMP22-180SH	

If it is not practical for teams to use one of the above motors or you find any other suitable motor you would like added to the list please post the motor and why you think it is a good fit on this forum thread: <https://junior.forum.robocup.org/t/robocupjunior-soccer-entry-stantard-kit-motor-whitelist-discussion/3516>